

Two HRS Core cognitive classification methods informed by HRS/HCAP

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Abstract

- **Importance:** population cognitive classification in HRS Core; dementia and MCI surveillance; limited assessment battery
- **Objective:** describe two HRS Core methods calibrated to HRS/HCAP (Core algorithm, PMM); compare with the HRS/HCAP Algorithm, consensus, Langa-Weir, and Hudomiet
- **Design:** 2016 HRS Core classification in HRS/HCAP participants; consensus-panel secondary comparator
 - Study sample: HRS/HCAP 2016 participants, N = 3,496
 - [Provenance: R/MS_MAIN/MS_001_analysis_cohort.R]
- **Methods:** Core algorithm; PMM
 - Primary comparator: HRS/HCAP Algorithm
 - Secondary comparators: (1) HCAP consensus; (2) Langa-Weir; (3) Hudomiet classification
 - Statistical analysis: dementia versus non-dementia kappa; 3-class quadratic weighted kappa (Normal, MCI, Dementia); all statistics other than frequencies use complex sampling weights
 - [Provenance: R/MS_MAIN/MS_010_results_objects.R]
- **Results:**
 - Dementia/non-dementia kappa
 - Core algorithm versus HRS/HCAP Algorithm: 0.352
 - PMM versus HRS/HCAP Algorithm: 0.429
 - 3-class kappa
 - Core algorithm versus HRS/HCAP Algorithm: 0.505
 - PMM versus HRS/HCAP Algorithm: 0.509
 - [Provenance: R_objects/A7_100_hcap_tables.rds; R/PMM_110_Comparison_of_Consensus_Langa_Weir.R]
- **Conclusions and relevance:** Two core classification approaches; agreement evidence rather than clinical validation; uncertainty in distinguishing MCI from normal cognition and dementia; limited precision of comparisons with consensus diagnoses because the consensus-panel subsample is small
 - [Provenance: analysis design; inference]

Key terms

- **HRS/HCAP Algorithm:** c.f. Manly, Jones et al. (2022)
- **Core algorithm:** HCAP algorithm adapted to the HRS Core data

- **PMM:** Profile mixture model using measures collected in the HRS Core, calibrated to the HRS/HCAP Algorithm and scored in the HRS/HCAP sample

Introduction

- Population cognitive classification from brief HRS Core assessments
 - Prevalence estimation; etiologic research; trial sampling; policy surveillance
 - Core assessment constraints; incomplete cognitive testing; proxy/informant information
- HRS/HCAP Algorithm as primary comparator
 - HCAP neuropsychological assessment; actuarial Normal/MCI/Dementia classification
 - HRS/HCAP Algorithm; primary HCAP-based comparator
 - Manly et al. (2022); Farron et al. (2025)
 - [Provenance: R/A7_100-comparison_of_diagnoses.R; existing manuscript materials]
- Study aim
 - Core algorithm and PMM; HRS/HCAP Algorithm primary comparator
 - Secondary comparators: HCAP consensus; Langa-Weir; Hudomiet
 - Classification margins; 3-class agreement; dementia/non-dementia agreement

Methods

Design and analytic samples

- Study sample: HRS/HCAP 2016 participants, N = 3,496
 - All HCAP participants; no Core cognitive-test requirement
 - Participants without Core cognitive testing classified from informant data (Jorm IQCODE): Core algorithm Jorm fallback; PMM Jorm model
 - Agreement-statistic samples: Core algorithm N = 3,496; PMM N = 3,496
 - [Provenance: R/MS_MAIN/MS_001_analysis_cohort.R; R_objects/A7_005_hrs16_merged.rds; R/A7_100-comparison_of_diagnoses.R; R/PMM_110_Comparison_of_Consensus_Langa_Weir.R]
- Consensus panel subset
 - Validation-subsample design; inverse-probability weighting to account for selection into the consensus panel
 - [Provenance: R/A7_100-comparison_of_diagnoses.R; R/PMM_110_Comparison_of_Consensus_Langa_Weir.R]

HRS Core classification approaches

- Two Core classification approaches
 - Core algorithm
 - Survey-weighted CFA; demographically normed cognitive score; cognitive/self-report/informant rules

- [Provenance: R/A7_050-mplus_model.R; R/A7_055-norm_factor_score.R; R/A7_075-implement_algorithm.R]
 - PMM
 - HRS/HCAP-calibrated probabilistic Normal/MCI/Dementia classification; combined cognition and Jorm models (i.e., two PMM models, one trained using cognition data, the other trained using informant data (Jorm IQCODE)).
 - *Limitation:* The HRS/HCAP IQCODE look back window is the past 10 years, the Core IQCODE has a look back window of 2 years.
 - [Provenance: R/PMM_103_calibration_model.R; R/PMM_110_Comparison_of_Consensus_Langa_Weir.R]
- Single-factor Core cognition model
 - Not enough items to separately score 5 domains matching Manly et al. (2022)
 - Initial 8-indicator, complex-survey model fit evaluated with WLSMV
 - Factor scores estimated with MLR, a probit link, and EAP scoring after fixing item parameters to the MLR model estimates
 - Factor-score normalization; age, sex, race/ethnicity, education covariates
 - [Provenance: R/A7_050-mplus_model.R; mplus_output/A7/model_1a_fixed/model_1a_fixed.inp; R/A7_055-norm_factor_score.R]
- Rule-based classification
 - Cognitive thresholds; self-reported concern; JORM for missing cognitive assessments
 - Normal, MCI, Dementia output
 - More detail in **Appendix 1**
 - [Provenance: R/A7_075-implement_algorithm.R]
- HRS/HCAP calibration
 - Known-class HRS/HCAP Algorithm classification; cognitive, functional, self-report, informant indicators
 - No latent variables for overall cognition, individual tests used
 - Calibration parameters fixed for scoring; scored in the same HRS/HCAP sample used for calibration (in-sample classification); not yet applied to the HRS Core sample outside HCAP
 - [Provenance: R/PMM_100_norming_npb.R; R/PMM_103_calibration_model.R]
- Class proportions (priors)

- Class-specific parameters estimated with HCAP16WGTR; MLR estimation, Bayes-rule posterior classification (plug-in rule; McLachlan, 1992)
- Class proportions fixed at the HCAP16WGTR-weighted HRS/HCAP Algorithm distribution, computed outside Mplus; Mplus KNOWNCLASS estimates class proportions from unweighted counts
 - Cognition model: Normal 69.4%, MCI 21.4%, Dementia 9.2%
 - Jorm model: Normal 31.7%, MCI 31.6%, Dementia 36.7%
- [Provenance: R/PMM_101_mplus_function.R; R/PMM_103_calibration_model.R; R/PMM_104_weight_check.R; mplus_output/pmm_103/pmm_hcap_103b.out; mplus_output/pmm_103_jorm/pmm_hcap_103b_jorm.out]
- Combined classification
 - Cognition-model probabilities; JORM-model probabilities when available
 - Probabilistic Normal/MCI/Dementia membership; replicated probability-weighted comparisons
 - More detail in **Appendix 2**
 - [Provenance: R/PMM_110_Comparison_of_Consensus_Langa_Weir.R]

Comparison classifications and estimands

- Primary comparator: HRS/HCAP Algorithm
 - HCAP population weights: HCAP16WGTR
 - [Provenance: R/A7_100-comparison_of_diagnoses.R]
- Secondary comparators
 - HCAP consensus diagnosis: normalized HCAP16WGTR x samplingP⁻¹ weights
 - Langa-Weir; Hudomiet classification
 - [Provenance: R/A0_017-recode_classification.R; R/A7_100-comparison_of_diagnoses.R]
- Agreement estimands
 - 3-class quadratic weighted kappa
 - Dementia versus non-dementia kappa
 - Marginal Normal/MCI/Dementia distributions
 - [Provenance: R/MS_MAIN/MS_010_results_objects.R]

Results

Analytic samples and classification margins

- Table 1: HRS/HCAP study sample characteristics
 - Dynamic counts and descriptive statistics

- [Provenance: R/MS_MAIN/MS_Tab_Fig_Apndx-030-Table-1.qmd]
- Table 2: weighted classification margins
 - HRS/HCAP Algorithm; Core algorithm; PMM; Langa-Weir; Hudomiet
 - HCAP16WGTR-weighted margins
 - HRS/HCAP Algorithm: Normal 69.4%, MCI 21.4%, Dementia 9.2%
 - Core algorithm: Normal 71.4%, MCI 20.5%, Dementia 8.0%
 - PMM: Normal 76.7%, MCI 13.6%, Dementia 9.7%
 - [Provenance: R/MS_MAIN/MS_010_results_objects.R]
- Figure 1: Core algorithm
 - Classification rules and participant counts
 - [Provenance: R/MS_MAIN/MS_Tab_Fig_Apndx-020-Figure-1.qmd]
- Figure 2: PMM class-probability distributions
 - Probability separation; uncertainty near class boundaries
 - [Provenance: R/PMM_112_Margins.R]

Agreement with HRS/HCAP Algorithm

- Core algorithm versus HRS/HCAP Algorithm: 3-class kappa 0.505; binary kappa 0.352
 - [Provenance: R_objects/A7_100_hcap_tables.rds]
- PMM versus HRS/HCAP Algorithm: 3-class kappa 0.509; binary kappa 0.429
 - [Provenance: R/PMM_110_Comparison_of_Consensus_Langa_Weir.R]
- Table 3: pairwise agreement, comparators, weight definitions
 - [Provenance: R/MS_MAIN/MS_010_results_objects.R]

Agreement with consensus and established HRS classifications

- Core algorithm, PMM, Langa-Weir, Hudomiet
 - Separate HCAP population and consensus-subsample estimands
 - Weight-dependent interpretation
 - [Provenance: R/MS_MAIN/MS_010_results_objects.R]
- Table 4: A7 pairwise three-class agreement matrix
 - [Provenance: R_objects/A7_100_hcap_tables.rds]

Discussion

- Two HRS Core classification approaches
 - Rule-based Core algorithm; probabilistic PMM classification
 - Same HRS/HCAP learning target; distinct measurement and classification assumptions

- [Provenance:analysis methods]
- Primary comparison
 - HRS/HCAP Algorithm; agreement and marginal-distribution evidence
 - [Provenance: Tables 2-4]
- Secondary comparisons
 - Consensus: small validation subset; inverse-probability weights
 - Langa-Weir and Hudomiet: established HRS comparators; different development targets
 - [Provenance: R/A7_100-comparison_of_diagnoses.R]
- Interpretation limits
 - Classification agreement; no clinical diagnostic validation outside HCAP consensus subset
 - MCI boundary; model-dependent and threshold-dependent assignments
 - PMM agreement with the HRS/HCAP Algorithm is in-sample (apparent error rate; McLachlan, 1992); expected to be optimistic relative to application in new samples
 - [Provenance: analysis design; inference; R/PMM_103_calibration_model.R]

**Two HRS Core cognitive classification methods informed by HRS/HCAP: Figures, tables,
and appendices**

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Table 1. Participant Characteristics

Characteristic	HRS/HCAP 2016 participants (N = 3,496)
Sex (N (%))	
Female	2,095 (59.9%)
Male	1,401 (40.1%)
Age in years (Mean (SD))	76.0 (7.6)
Race (N (%))	
White	2,755 (78.8%)
Black	563 (16.1%)
All other groups	177 (5.1%)
Not collected	1 (0.0%)
Hispanic origin (N (%))	
Not Hispanic	3,111 (89.0%)
Hispanic	383 (11.0%)
Not obtained	2 (0.1%)
Years of education (Mean (SD))	12.7 (3.2)
HRS/HCAP Classification (N (%))	
Normal	2,313 (66.2%)
Mild cognitive impairment	808 (23.1%)
Dementia	375 (10.7%)

Table 1. Participant Characteristics

Characteristic	HRS/HCAP 2016 participants (N = 3,496)
Langa-Weir Classification (N (%))	
Normal	2,432 (69.6%)
Cognitively impaired, not demented	731 (20.9%)
Dementia	333 (9.5%)
Jorm IQCODE score from Core (Mean (SD))	3.8 (0.8)
Jorm IQCODE score from Core (missing)	3,283

Source: R/MS_MAIN/MS_001_analysis_cohort.R;
R_objects/A7_005_hrs16_merged.rds.

Jorm IQCODE was collected in the HRS Core interview, not HCAP, and only from participants who did not complete Core cognitive testing.

Notes: The study sample includes all 3,496 HRS/HCAP 2016 participants, including participants without Core cognitive testing who were classified from informant (Jorm IQCODE) data. Race and ethnicity data were gathered via self-report at the time of first interview and are considered to be markers of exposure to evolving systems of racism, not as a proxy for genetic variation or any other biological variable. Race was self-selected by participants at the time of the first interview from a list of options defined by the 2000 US Census criteria. Other race includes a pooled group of participants who identified as American Indian or Alaska Native, Asian, Native Hawaiian or Pacific Islander, or another self-described race, consolidated due to small sample sizes and risk of identification.

Table 2. HCAP classification margins by method

Method	Normal		MCI		Dementia	
	N	Weighted %	N	Weighted %	N	Weighted %
HRS/HCAP Algorithm	2,313	69.4%	808	21.4%	375	9.2%
Core algorithm	2,453	71.4%	735	20.5%	308	8.0%
PMM	2,557	76.7%	568	13.6%	371	9.7%
Secondary comparators						
Langa-Weir	2,432	74.9%	731	17.3%	333	7.9%
Hudomiet	2,326	71.1%	865	21.0%	286	7.2%

Note. N values are unweighted HRS/HCAP participant counts. Percentages use HCAP16WGTR; PMM counts and percentages use the non-replicated combined cognition/Jorm model sample.

References. Manly, J. J., Jones, R. N., Langa, K. M., Ryan, L. H., Levine, D. A., McCammon, R., Heeringa, S. G., & Weir, D. (2022). Estimating the prevalence of dementia and mild cognitive impairment in the US: The 2016 Health and Retirement Study Harmonized Cognitive Assessment Protocol Project. *JAMA Neurology*, 79(12), 1242-1249. <https://doi.org/10.1001/jamaneurol.2022.3543>

Langa, K. M., Weir, D. R., Kabeto, M., & Sonnega, A. (2023). Langa-Weir Classification of Cognitive Function (1995-2020). https://hrsdata.isr.umich.edu/sites/default/files/documentation/data-descriptions/1695907706/Data_Description_Langa_Weir_Classifications2020_V2.pdf

Hudomiet, P., Hurd, M. D., & Rohwedder, S. (2022). Trends in inequalities in the prevalence of dementia in the United States. *Proceedings of the National Academy of Sciences*, 119(46), e2212205119. <https://doi.org/10.1073/pnas.2212205119>

Table 3. Agreement among Core classifications and comparison diagnoses

Classification 1	Classification 2	Three-class quadratic weighted kappa	Dementia versus non-dementia kappa
Core Algorithm	HRS/HCAP Algorithm	0.505	0.352
PMM	HRS/HCAP Algorithm	0.509	0.429
Langa-Weir	HRS/HCAP Algorithm	0.510	0.407
Hudomiet	HRS/HCAP Algorithm	0.551	0.457
HCAP Consensus	HRS/HCAP Algorithm	0.728	0.536
Core Algorithm	Langa-Weir	0.561	0.444
PMM	Langa-Weir	0.575	0.480
Hudomiet	Langa-Weir	0.717	0.670
HCAP Consensus	Langa-Weir	0.099	0.736
Core Algorithm	Hudomiet	0.603	0.474
PMM	Hudomiet	0.623	0.529
HCAP Consensus	Hudomiet	0.396	0.806
Core Algorithm	HCAP Consensus	0.284	0.675
PMM	HCAP Consensus	0.486	0.464

Source: A7 rows: R_objects/A7_100_hcap_tables.rds, generated by R/A7_100-comparison_of_diagnoses.R. PMM rows: R/PMM_110_Comparison_of_Consensus_Langa_Weir.R; PMM-Hudomiet was calculated from the same saved probabilities and A7 comparison data. Three-class values are quadratic weighted kappas; binary values are kappas for dementia versus non-dementia. Consensus comparisons use normalized HCAP16WGTR / samplingP weights; other comparisons use HCAP16WGTR.

Table 4. Pairwise three-class agreement among HCAP comparison methods

method	Consensus diagnosis	Algorithmic diagnosis	Langa-Weir	HRS classification model	Hudomiet
Consensus diagnosis	1.000	0.728	0.099	0.284	0.396
Algorithmic diagnosis	0.728	1.000	0.510	0.505	0.551
Langa-Weir	0.099	0.510	1.000	0.561	0.717
HRS classification model	0.284	0.505	0.561	1.000	0.603
Hudomiet	0.396	0.551	0.717	0.603	1.000

Source: Source: R_objects/A7_100_hcap_tables.rds, generated by R/A7_100-comparison_of_diagnoses.R. Entries are quadratic weighted kappas. Consensus comparisons use normalized HCAP16WGTR / samplingP weights; other comparisons use HCAP16WGTR.

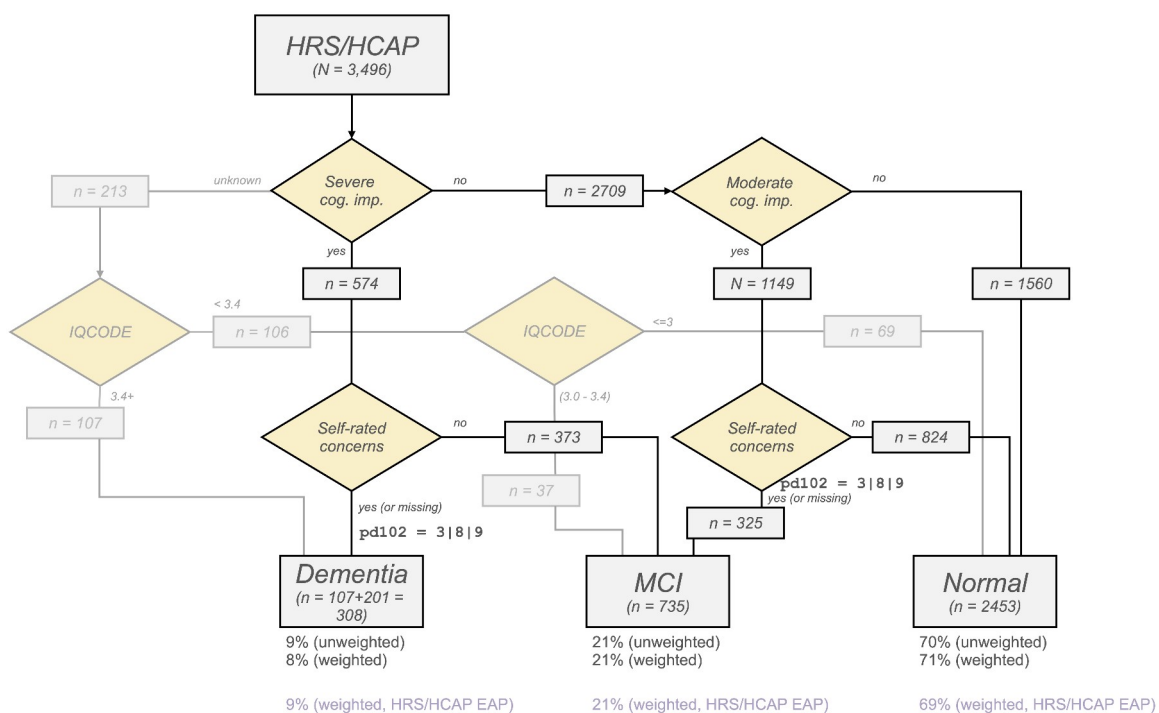


Figure 1: Core algorithm

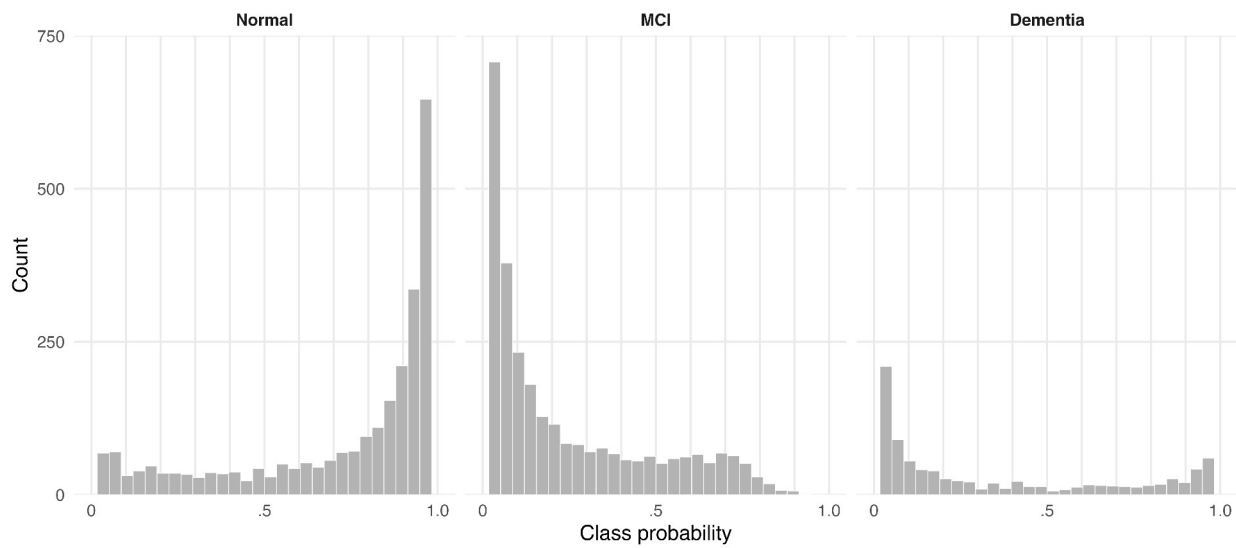


Figure 2: PMM class probabilities in HRS/HCAP

Thanks for reading.

Appendix 1. Development of the Core algorithm

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- **Core cognition model**
 - Eight HRS Core cognitive indicators; single-factor, complex-survey model
 - Initial model fit evaluated with WLSMV
 - Fixed-parameter factor scores estimated with MLR, a probit link, and EAP scoring
 - [Provenance: R/A7_050-mplus_model.R;
mplus_output/A7/model_1a_fixed/model_1a_fixed.inp]
- **Score normalization**
 - Age, sex, race/ethnicity, education, and interaction covariates
 - Blom transformation; demographically adjusted residual T score
 - [Provenance: R/A7_055-norm_factor_score.R]
- **Rule-based classification**
 - Cognitive impairment; function; self-reported concerns; JORM for non-completers
 - Normal, MCI, Dementia classifications
 - [Provenance: R/A7_075-implement_algorithm.R]
- **Relationship to HRS/HCAP Algorithm**
 - HCAP-derived learning target; Core-measure operationalization
 - [Provenance: R/A7_100-comparison_of_diagnoses.R]

Appendix 2. PMM and logistic regression

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PMM and logistic regression can both classify people into Normal, MCI, or Dementia. The difference is what they model.

Logistic regression

Logistic regression starts with a known outcome, Y , and predicts it from observed predictors, X :

$$P(Y = \text{Dementia} | X)$$

For example, it could predict dementia from word recall, serial sevens, ADLs, and Jorm score. It estimates how each predictor changes the log-odds of each outcome category. In a multinomial logistic model, the usual form is:

$$\log \frac{P(Y = k | X)}{P(Y = \text{Normal} | X)} = \beta_{0k} + \beta_{1k} X_1 + \cdots + \beta_{pk} X_p$$

The main modeling decision is the regression equation: which predictors, interactions, nonlinear terms, and coefficients to include.

PMM in this project

The PMM takes a generative approach. It asks: *If a person belongs to the Normal, MCI, or Dementia class, what pattern of cognitive-test, functional, self-report, and informant responses would we expect?* PMM models:

$$P(X | C = k)$$

for each cognitive class $C = k$, along with the class prevalence $P(C = k)$. It then uses Bayes' rule to obtain the classification probability:

$$P(C = k | X) = \frac{P(X | C = k) P(C = k)}{\sum_j P(X | C = j) P(C = j)}$$

So, instead of directly predicting diagnosis from scores, it describes the characteristic response distribution within each diagnosis class and asks which class best accounts for a person's observed pattern.

For the final PMM, the cognition model uses the individual test scores, functional items, and self-report item directly. It estimates class probabilities from that whole response pattern. The separate Jorm PMM does the analogous job for people with informant data rather than cognitive testing.

Practical distinctions

- Logistic regression: “Given these scores, what is the probability of Dementia?”
- PMM: “Which diagnosis class most plausibly generated this pattern of scores and reports?”
- Logistic regression is discriminative: it models $P(Y|X)$ directly.
- PMM is generative: it models $P(X|C)$, then derives $P(C|X)$.
- Logistic regression commonly treats predictors as explanatory variables with regression coefficients.
- PMM treats the observed responses as indicators of an underlying class-specific response profile.
- Logistic regression needs explicit interactions or nonlinear terms when the meaning of one score depends on another.
- A PMM can represent distinct multivariate profiles through class-specific item distributions, although it still relies on assumptions about within-class dependence.

In this project, the important wrinkle is that the PMM is not purely exploratory latent class analysis. It is calibrated using the HRS/HCAP Algorithm classification as the known class in the HCAP sample, then scores the Core measures of the same HCAP participants with the calibrated parameters (including the class proportions) held fixed. That makes it a supervised, model-based classifier: conceptually closer to generative discriminant analysis than to an unsupervised profiling exercise.

A concise manuscript description would be:

The PMM is a supervised, generative classifier that estimates the probability of each HRS/HCAP Algorithm class from the joint pattern of observed cognitive, functional, self-report, or informant responses.

Appendix 3: Comparisons with other study classifications

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